

PROFESSIONAL SUMMARY

- Highly motivated professional engineer with 7+ years of experience in hardware testing, failure analysis, and DOE-driven process improvement across Semiconductor and Electronics manufacturing.
- Gained strong technical problem-solving experience in Semiconductor, Telecom & Medical Electronics Manufacturing Organization.

PROFESSIONAL EXPERIENCE

07-03-2024 to present

DOE and Chamber Validation Engineer
Foxconn Technology pvt ltd || Bangalore ||

- To identify root cause for DUT performance in each configuration or build that failure to achieve its intended RF performance, and providing the input for PD and Hardware team about the failure reasons and root causes to overcome the issues in each NPI stages (P1/P2/EVT/CRB & DVT).
- Optimized antenna matching by routing alternative RF paths to the Tuner IC via baseband-controlled switching, improving antenna resonance and radiation efficiency across LTE, 5G-NR, and Wi-Fi bands.
- Executed Design of Experiments involving housing trim, flex cut, copper trace addition, BOM changes, spring modification, and ground-path optimization to mitigate detuning and improve antenna Performance.
- Performed OTA validation in anechoic chambers including Active TRP/TIS and Passive RF parameters (S11/Return Loss, VSWR) through DOE-based impedance-matching optimization.
- Responsible for analyzing testing data across wide & short distributions to ensure RF TRP meets define specification, conduct configuration wise validation to identify the complex issues by using BOX plot for quick visualization and root – cause identification of performance.
- Conducted cSAR testing to evaluate SAR compliance and antenna behavior across multiple tuner configurations.
- Performed reverberation chamber testing using CMW-based signal injection to the DUT to evaluate signaling performance and antenna robustness in noisy and multipath environments.
- Performed Phantom and user-blocking tests with multiple coverage scenarios (bottom, side, top blockage) to assess real-world antenna performance degradation.
- Used Spectrum Analyzer to measure RX path noise levels and harmonic noise from the DUT, identifying interference sources and resolving RF desense issues affecting receiver sensitivity.
- Validated RF Port Isolation to ensure stable coexistence between LTE / NR / Wi-Fi / BT technologies and to prevent coupling-induced interference issues.
- Reviewed X-ray, CT scan, VHX microscope, and functional tester analysis to identify solder joint cracks, voids, or pad lifting contributing to intermittent RF failures.

08-06-2023 to 09-02-2024

Lab Support Engineer
Micron Technology-OTSI || Hyderabad ||

- ESD measurements of all LAB equipment and maintain ESD measurement log in internal portal.
- Implemented and support RFID-based Inventory tracking systems for lab assets, ensuring accurate real-time monitoring and traceability of high-value components and test equipment's
- Performed lab test bench setup for engineering validation by connecting test device ,RF Cables and measurement instruments ensuring stable test environment
- Support hardware bring-up and functional validation of electronics boards, signal paths and basic functionality before test.
- Configured and verify lab measurement equipment such as Oscilloscope, Spectrum Analyzer & Signal Generator to ensure accurate hardware validation results.
- Prepared and maintained calibration certificate and reports and ensuring up-to-date documentation was readily available for audit and quality assurance.
- Generate Monthly lab status report covering tool availability, calibration completion, material shortages and overall lab readiness.

21-02-2022 to 21-03-2023

Process Engineer
Avary Holding India PVT LTD || Chennai ||

- Work with various teams to develop standard work instruction to diagnose failures in Assembly Line.
- Perform Failure analysis on FBC & RF related components to identify rejection points and process defects in production environment.
- Analyzes the Products Quality by understanding the Failure mode through ICT/Functional testing and determine the root causes for failure
- Generate failure analysis report with detailed findings, root cause identification, and corrective action recommendations.
- Monitored assembly equipment's and test stations to identify RF & Functional connectivity issues, Solder defects and signal path Failures.
- Supported yield improvement activities by analyzing defect trends, test failures and process variation in assembly operations.
- Performed time study and line balancing to improve manufacturing efficiency and reduce production delays in assembly line.
- SOP and action report presentation skills.

10-08-2020 to 14-02-2022

Calibration/Testing Engineer

Breath-X Technology pvt ltd, a Canary Global Inc|| Bangalore ||

- Perform accurate calibration and record Test performed by equipment's (i.e., Humidity & pH-Meter, SMT rework Station, Soldering Station, DMM, etc.,)
- Understanding of PLC Programming ladder logic and all aspects of inputs and outputs of the system.
- Provide guidelines to equipment users and other technicians in the proper operation of test and measurement equipment's.
- Assist in development of new tests or standards to upgrade quality testing capabilities.
- Providing ongoing quality engineering support throughout the product life cycle, including risk management, management changes and CAPA investigations.

10-09-2018 to 04-06-2020

Testing/Analysis Engineer Trainee (NATS)

Flextronics Technology (India) pvt ltd || Sricity, Andhra Pradesh ||

- Testing electronics to confirm working condition.
- Perform maintenance and Calibrations checks on test systems.
- Effect repairs to assigned equipment by following established Electrostatic Discharge (ESD) and safety standards and procedures.
- Practical knowledge of soldering /De-soldering electronics components on complex PCB package such as QFN, QFP, BGA, PGA, SOP, SOT,0402,0603,0805 etc.,
- Track personal maintenance efficiency and rework rate, and rework feedback of defect and assist the improvement.

SKILLS

Leadership Skills

- Responsible for improving processes, maintaining the quality of the product, floor troubleshooting and training associates.

Computer Skills

- Microsoft: Office, Excel & PowerPoint.

Root Cause Analysis

- Received training in Root Cause Assessment and Counter-measure training.
- Conducted Root cause analysis for customer returns and implemented counter-measure.

EDUCATION

Bachelors in Electronics & Communication Engineering (RMK Engineering College) Chennai -2015 to 2018

Diploma in Electronics & Communication Engineering (Sri Durga-devi Polytechnic College) Chennai. - 2012 to 2015

SSLC (St. Mary's Matriculation School) Chennai - 2011 to 2012

TECHNICAL SKILLS

- Effective Trouble shooting skills and analyzing circuits.
- Test chamber Audit & verification
- SMT rework & component level debugging
- cSAR, Reverb & Chamber Testing.

CERTIFICATION

- Certified IN ISO-9001:2015 (Quality Management System)
- PCB Design using Easy-EDA (Single & Double sides PCB).
- CCNA Routing and Switching.

PERSONAL DETAILS

DATE OF BIRTH : 06/05/1997

GENDRE : Male

NATIONALITY : Indian

LANGUAGES KNOWN : English, Tamil, Telugu & Kannada

PERMENENT ADDRESS : S/O B Mohan Kumar, Flat no 11, AVM Nagar, Tada(M), Tirupati(D),524401.

DECLARATION

I hereby insist that the information provided is accurate to the best of my facts and belief.

Date:

Place:

LOKESH M